

Aditya Joshi

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EDUCATION

Savitribai Phule Pune University

Nov 2022 – May 2026

BE in Computer Engineering with Honours in DS and ML

- CGPA: 7.9/10

EXPERIENCE

AI/ML Intern | Hilwitz Solutions Pvt. Ltd.

Nov 2025 – Apr 2026

- Completed a six-month internship focused on AI/ML development, building and deploying machine learning solutions.

Intern – GenAI | Maharashtra Knowledge Corporation Ltd. (MKCL)

Mar 2025 – Sept 2025

- Built a Flask-based API and Vue.js frontend for an Employee Directory System with authentication and role-based access.
- Developed a multilingual Graph RAG chatbot running on a locally hosted LLM with Neo4j, capable of answering queries on government schemes datasets in both English and Marathi.
- Strengthened expertise in GenAI applications, multilingual NLP, knowledge graphs, and full-stack development.

Backend Development Workshop | Netcamp Solutions Pvt. Ltd.

Jun 2024 – Aug 2024

- Developed and optimized RESTful APIs using Python (Flask) to enhance system efficiency.
- Designed and implemented an SQL database schema to ensure smooth data management and retrieval.
- Worked on a JavaScript-based front-end project, improving UI/UX in collaboration with peers.

PUBLICATIONS

Understanding Quantum Computers

Jan 2024

Aditya Joshi, Pratham Dixit, Avdhut Pithe — MMCOE, Yearly Magazine

PROJECTS

Government Schemes Graph RAG Chatbot

github.com/adi-21-4/graphbased-chatbot

- Built a full-stack, voice-enabled Retrieval-Augmented Generation chatbot answering government scheme queries in English and Marathi.
- Used a Neo4j knowledge graph with a local LLM to translate natural-language questions into Cypher queries for relationship-based retrieval.
- Implemented speech-to-text via the Web Speech API and offline text-to-speech with pyttsx3; frontend in Vue 3 + Tailwind CSS, backend in Python (Flask).

AI Powered Chat-bot

github.com/adi-21-4/chatbot

- Developed an AI-driven chatbot to handle basic student queries related to faculty and administration.
- Used JavaScript for the frontend, with a JSON-based database for storing responses.
- Integrated tokenization to improve query understanding and accuracy.

Flask API Backend

github.com/adi-21-4/api-backend-with-flask

- Built a Flask-based REST API implementing CRUD operations, file upload, and retrieval against a MySQL database.
- Organized the codebase into controller and model layers, with authentication handling.

Movie Recommendation System

github.com/adi-21-4/movie-recommendation-system

- Built a machine learning-based recommendation system using Python (Pandas, Scikit-Learn).
- Utilized collaborative filtering algorithms to suggest movies based on user preferences.
- Designed an interactive web interface using Flask and HTML/CSS.

TECHNOLOGIES

Languages: Python, C++, Java, SQL, JavaScript, Kotlin

Technologies: .NET, Microsoft SQL Server, MySQL, Neo4j, LangChain, Flask, Vue.js, Scikit-Learn, Postman

PUBLICATIONS (CONTINUED)

Conversational Chatbot for Government Schemes Using Neo4j Knowledge Graph and Multilingual Voice Interface

Unpublished

Aditya Joshi, Anshul Dhore, Girija Dahiphale, Pratham Dixit, Snehal Karad

GraphRAG-powered chatbot for Indian government schemes, combining a Neo4j knowledge graph with a locally deployed LLM and a trilingual (English/Hindi/Marathi) voice interface. 92.4% intent accuracy, 95.2% Top-3 retrieval accuracy across 512 evaluated queries.